On Sampled Metrics for Item Recommendation

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- Item recommendation algorithms are evaluated by metrics that compare the position of truly relevant items among the recommended items.
- Different methods for comparing an algorithm : AUC, Precision, Recall, Average Precision, NDCG.

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- How can we improve the quality of estimation when we need to use a sample set?

Evaluating an algorithm

n input items $\rightarrow$ Algorithm A ranks for instance x

Average metric reported

$$\frac{1}{|D|}\sum_{x\in D} M(R(A,x))$$

where $D = \{x_1, x_2, ...\}$ for multiple instances and $R(A,x)$ are predicted ranks for instance x.

Example: $R(A,x) = \{3,5\}$

What is Sampling and Consistency?

Sample Set

All relevant + m random irrelevant from total n items.

Consistency

$$\frac{1}{|D|}\sum_{x\in D} M(R(A,x)) > \frac{1}{|D|}\sum_{x\in D} M(R(B,x))$$

$$\iff$$

$$E[\frac{1}{|D|}\sum_{x\in D} M(R(A,x))] > E[\frac{1}{|D|}\sum_{x\in D} M(R(B,x))]$$

- If a metric is inconsistent, then measuring M on a subsample is not a good indicator of the true performance of M.

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- $\text{Recall}(r)_k = \delta(r \leq k)$
- $\text{AUC}(r)_k = \frac{n-r}{n-1}$
- $\text{NDCG}(r)_n = \delta(r \leq k)\frac{1}{log_2(r+1)}$

Simplified Metrics: Example

Example: Reccomenders A, B and C, $n = 10000$, $|D| = 5$ instances with 1 relevant item in each instance.

- Recommender A : never good or terrible, ranks the relevant item at position 100 each time.
- Recommender B : ranks relevant item at position 40 for 2 instances.
- Recommender C : ranks relevant item at position 2 in one of the instance, but never able to achieve a good rank for other 4 instances.

Simplified Metrics: Example

	Predicted Ranks	AUC	AP	NDCG	Recall@10
A	100, 100, 100, 100, 100	**0.990**	0.010	0.150	0.000
B	40, 40, 8437, 9266, 4482	0.555	0.010	0.122	0.000
C	212, 2, 743, 5342, 1548	0.843	**0.101**	**0.208**	**0.200**

Table: Toy Example of evaluating three recommenders A, B, and C on five instances.

- Equally cares about ranks in AUC
- In top heavy metrics C performs better

Simplified Metrics: Example

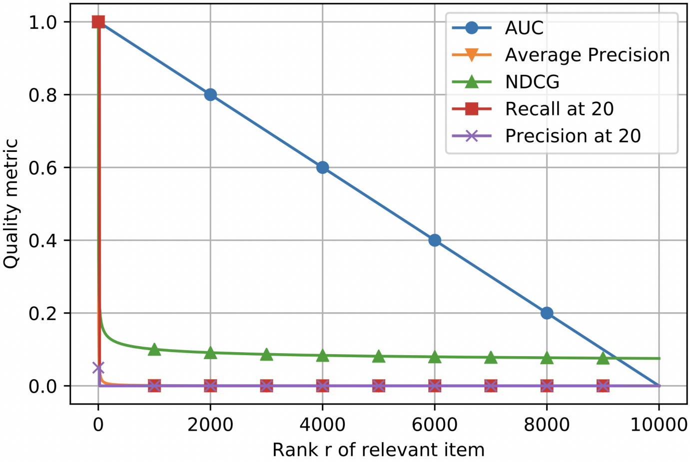

Figure: Metrics over the whole set of 10000 items.

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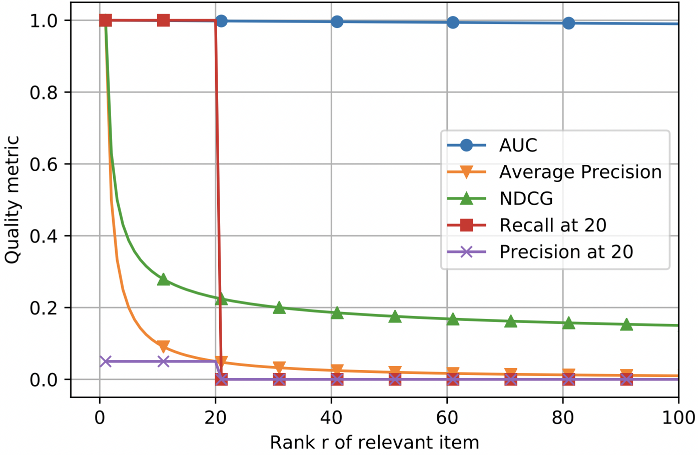

Figure: Contributions of the top 100 ranks.

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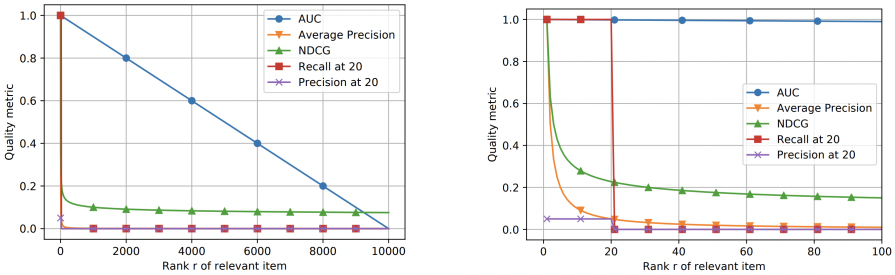

Figure: Visualization of metric vs. predicted rank for $n = 10000$. The left side shows the metrics over the whole set of 10000 items. The right side zooms onto the contributions of the top 100 ranks. Side by side comparsion.

- All metrics besides AUC are top heavy and almost completely ignore the tail.

Understanding the problem

- Previous example: A, B, C recommenders. $n = 10000, |D| = 5$, A : 100 each time, B : 40 twice, C : 2 once
- Sampled set: 1 relevant item + $m = 99$ random irrelevant items
- Repeated 1000 times and average and standard deviation is computed.

	Predicted Ranks	AUC	AP	NDCG	Recall@10
A	100, 100, 100, 100, 100	**0.990**±0.004	**0.630**±0.129	**0.724**±0.097	**1.000**±0.000
B	40, 40, 8437, 9266, 4482	0.555±0.014	0.336±0.073	0.444±0.054	0.400±0.000
C	212, 2, 743, 5342, 1548	0.843±0.014	0.325±0.050	0.460±0.039	0.567±0.092

Figure: Sampled evaluation for the recommenders A, B, C.

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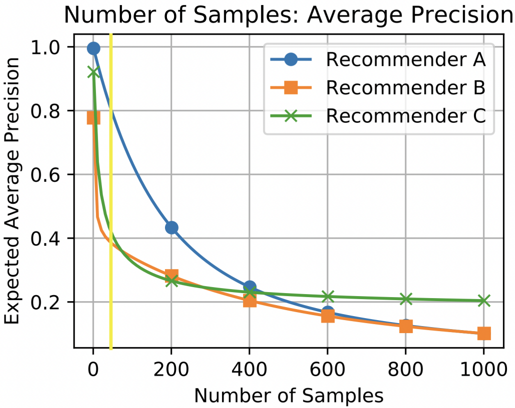

Figure: Expected Sampling average Precision for the running example.

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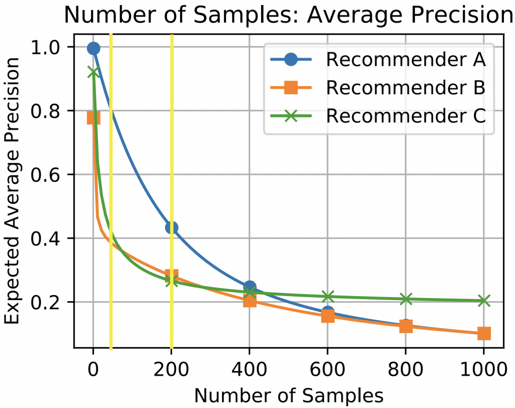

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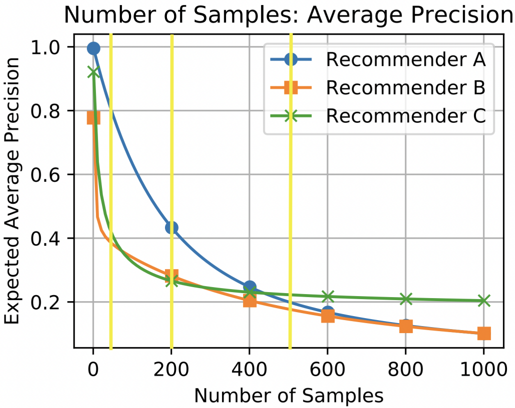

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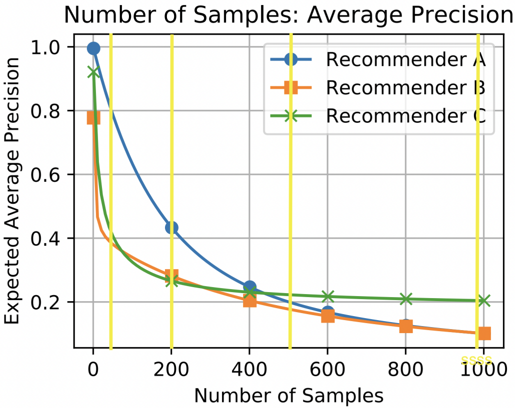

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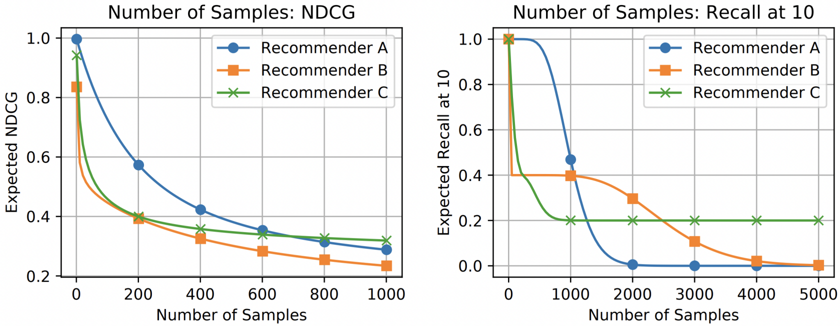

Figure: Expected Sampling NDCG and Recall@10 for the running example.

Understanding the problem

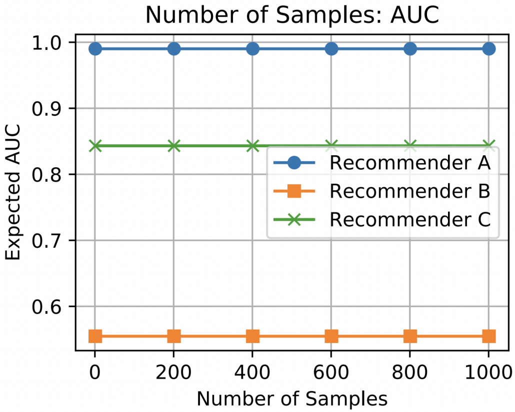

Figure: Expected Sampling AUC for the running example.

Expectation on metrics

- The rank $\tilde{r}$ (measured rank on sample) obtained from the sampling process follows a binomial distribution.

$$\tilde{r} \sim B(m, \frac{r-1}{n-1}) + 1$$

- AUC is linear in r (true rank of unique relevant item). This gives us the expectation of AUC on $\tilde{r}$ as $\mathrm{AUC}_n(r)$ which means that measurements created by sampling are unbiased estimators of the exact AUC.
- Expectation of Recall and Precision is a cumulative distribution of binomial distribution.

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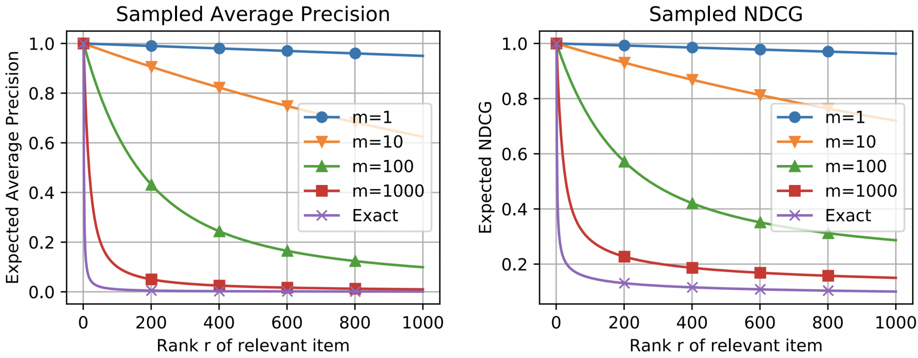

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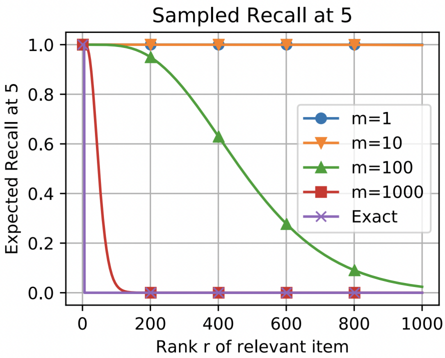

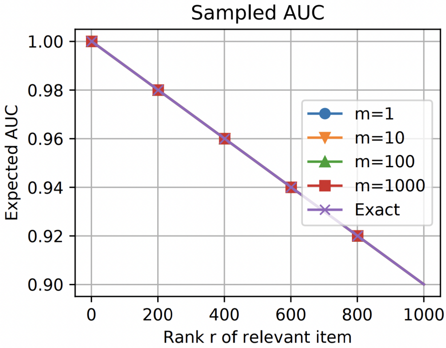

Metrics for small sample size

- In this case $\tilde{r} \in \{1, 2\}$ and for any metric M, $E[M(\tilde{r})] = p(\tilde{r} = 1)M(1) + (1 - p(\tilde{r} = 1))M(2)$

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- This gives a linear expression in r with its coefficient as $M(2) - M(1)$.
- For metrics which cannot differentiate between first and second position (Precision, Recall @ $k \geq 2$), M is constant hence not useful.
- For a reasonable metric, the coefficient of r should be negative.
- This shows that any sensible metric gives the same qualitative statement for $m = 1$.

Corrected Metrics

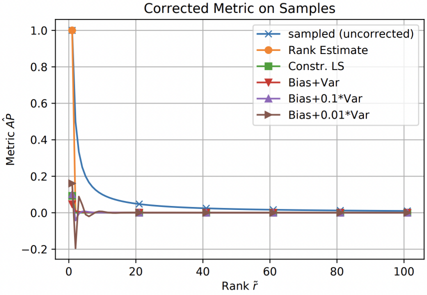
Can we design a sampled metric $\hat{M}$, a function from $\{1, ..., m+1\}$ to $\mathbf{R}$ such that $\hat{M}(\tilde{r})$ provides a good estimate of $M(r)$?

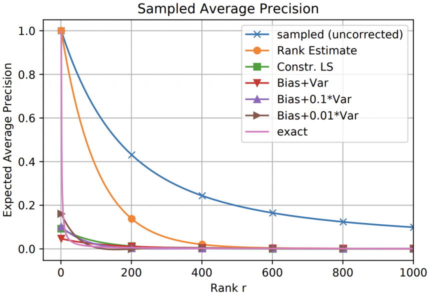
Correction methods:

- Unbiased Estimator of the Rank
- Minimal Bias Estimator
- Bias-Variance Trade-off

Corrected Metrics

Example





Experiments

- Do recommender algorithms create different ranking distributions?
- Are results from sampled metrics and exact metrics inconsistent?
- Can corrections help to get more reliable results?

Dataset: Binarized Movielens 1M, split, sampling size ($m = 100$) and metrics (Recall@10 and NDCG@10).

Experiments

Figure: Evaluation of three recommenders (X, Y and Z) on the Movielens dataset. Sampled metrics are inconsistent with the exact metrics. Corrected metrics produce the correct relative ordering in expectation.

			Sampled	Sampled with Correction					
	Recommender	Exact	(uncorrected)	Rank Estimate	CLS	BV 1	BV 0.1	BV 0.01	BV 0.001
Recall	X	7.60	66.19±0.25	17.46±0.32	8.52±0.16	4.71±0.07	6.49±0.25	7.18±0.59	7.32±1.17
	Y	8.84	56.51±0.22	17.26±0.28	8.42±0.14	4.60±0.07	6.95±0.21	8.18±0.51	8.54±1.07
	Z	9.42	54.20±0.22	18.67±0.32	9.10±0.15	4.97±0.07	7.44±0.24	8.63±0.59	9.08±1.20
NDCG	X	3.76	39.21±0.20	17.46±0.32	3.99±0.07	2.16±0.04	3.00±0.12	3.34±0.32	3.41±0.71
	Y	4.59	34.82±0.16	17.26±0.28	3.94±0.06	2.12±0.03	3.24±0.10	3.85±0.28	4.03±0.66
	Z	4.79	35.34±0.16	18.67±0.32	4.27±0.07	2.29±0.04	3.46±0.12	4.05±0.32	4.31±0.74
AP	X	3.75	32.55±0.21	18.12±0.31	3.58±0.06	2.44±0.03	3.13±0.09	3.37±0.21	3.42±0.49
	Y	4.32	30.01±0.20	17.81±0.28	3.54±0.06	2.32±0.03	3.19±0.07	3.62±0.18	3.73±0.45
	Z	4.44	30.71±0.21	19.20±0.31	3.82±0.06	2.45±0.03	3.38±0.08	3.79±0.21	3.97±0.51
AUC	X	89.13	89.12±0.04	89.24±0.04	89.12±0.04	88.36±0.04	89.04±0.04	89.11±0.04	89.12±0.04
	Y	85.33	85.33±0.04	85.48±0.04	85.32±0.04	84.63±0.04	85.26±0.04	85.32±0.04	85.32±0.04
	Z	74.73	75.04±0.23	75.24±0.20	75.04±0.21	74.51±0.23	75.02±0.20	75.02±0.24	75.02±0.23

Suggestions and Conclusion

Conclusion: Most metrics are inconsistent under sampling and can lead to false discoveries.

Suggestions:

- Sampling should be avoided during evaluation.
- Low-variance can give false confidence in wrong conclusion.
- If not avoidable: corrected metrics can reduce bias.

Thank You!